

BigCodeBench/83 Evaluations

Task Description

Create a Flask application configured to send emails using Flask-Mail with:

- Dynamic SMTP configuration based on parameters
 - A route to send test email
 - Return Flask application instance
 - Parameters: smtp_server, smtp_port, smtp_user, smtp_password, template_folder
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Model 1 Evaluation

model1_efficiency

Total steps: 6

- Step 1: Create Flask instance - **Essential**
- Step 2: Configure SMTP settings - **Essential**
- Step 3: Configure template folder - **Essential**
- Step 4: Initialize Mail extension - **Essential**
- Step 5: Define test email route - **Essential**
- Step 6: Return Flask app - **Essential**

model1_logic

Step Coherence:

- Step 1: Starting point
- Step 2: Coherent - configures after creation
- Step 3: Coherent - notes template_folder in Flask constructor
- Step 4: Coherent - Mail initialization after config
- Step 5: Coherent - adds required route
- Step 6: Coherent - returns as specified

Internal Consistency: Minor issue - Step 3 mentions configuring template_folder after Step 1 already created Flask instance

model1_completeness

Problem Coverage:

- Flask app creation: Yes
- SMTP configuration: Yes
- Template folder: Yes
- Mail initialization: Yes
- Test route: Yes
- Return app: Yes

Factual Correctness: Correct understanding of Flask-Mail configuration

Constraint and edge case handling:

- Mentions TLS for port 587
 - Query parameter for recipient
 - Default sender configuration
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Model 2 Evaluation

model2_efficiency

Total steps: 6

- Step 1: Import modules - **Non-essential** (already in starter code)
- Step 2: Define function and create app - **Essential**
- Step 3: Configure SMTP settings - **Essential**
- Step 4: Initialize Mail - **Essential**
- Step 5: Define test route - **Essential**
- Step 6: Return app - **Essential**

model2_logic

Step Coherence:

- Step 1: Starting point (though redundant)
- Step 2: Coherent - creates app with template_folder
- Step 3: Coherent - configures SMTP
- Step 4: Coherent - Mail initialization
- Step 5: Coherent - adds route
- Step 6: Coherent - returns app

Internal Consistency: No contradictions, mentions JSON response

model2_completeness

Problem Coverage:

- Flask app creation: Yes
- SMTP configuration: Yes (TLS/SSL mentioned)
- Template folder: Yes
- Mail initialization: Yes
- Test route: Yes (with error handling)
- Return app: Yes

Factual Correctness: Correct configuration approach

Constraint and edge case handling:

- Error handling in route
 - JSON response format
 - TLS enabled, SSL disabled by default
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Model 3 Evaluation

model3_efficiency

Total steps: 8

- Step 1: Problem understanding - **Essential**
- Step 2: Initialize Flask app - **Essential**
- Step 3: Configure SMTP - **Essential**
- Step 4: Handle encryption - **Essential**
- Step 5: Initialize Mail - **Essential**
- Step 6: Define route - **Essential**
- Step 7: Send email logic - **Essential**
- Step 8: Return app - **Essential**

model3_logic

Step Coherence:

- Step 1: Starting point
- Step 2: Coherent - app creation
- Step 3: Coherent - SMTP config

- Step 4: Coherent - security settings based on port
- Step 5: Coherent - Mail binding
- Step 6: Coherent - route definition
- Step 7: Coherent - email sending details
- Step 8: Coherent - final return

Internal Consistency: No contradictions, thorough approach

model3_completeness

Problem Coverage:

- Flask app creation: Yes
- SMTP configuration: Yes
- Template folder: Yes
- Mail initialization: Yes
- Test route: Yes
- Return app: Yes

Factual Correctness: Correct port-based encryption logic (465=SSL, else TLS)

Constraint and edge case handling:

- Smart port-based encryption selection
 - Hardcoded test values
 - Default sender configuration
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Model 4 Evaluation

model4_efficiency

Total steps: 6

- Step 1: Understand requirements - **Essential**
- Step 2: Design app structure - **Essential**
- Step 3: Handle security - **Essential**
- Step 4: Define email route - **Essential**
- Step 5: Implement sending - **Essential**
- Step 6: Return app - **Essential**

model4_logic

Step Coherence:

- Step 1: Starting point
- Step 2: Coherent - structural planning
- Step 3: Coherent - security considerations
- Step 4: Coherent - route planning
- Step 5: Coherent - implementation details
- Step 6: Coherent - return requirement

Internal Consistency: No contradictions, high-level design approach

model4_completeness

Problem Coverage:

- Flask app creation: Yes
- SMTP configuration: Yes
- Template folder: Yes
- Mail initialization: Yes
- Test route: Yes
- Return app: Yes

Factual Correctness: Correct security practices (SSL for 465, TLS otherwise)

Constraint and edge case handling:

- Port-based security selection
 - Fixed test email details
 - Common practices mentioned
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Model 5 Evaluation

model5_efficiency

Total steps: 4

- Step 1: Task understanding and imports - **Essential**
- Step 2: App initialization and config - **Essential**
- Step 3: Mail init and route definition - **Essential**
- Step 4: Return app - **Essential**

model5_logic

Step Coherence:

- Step 1: Starting point
- Step 2: Coherent - setup phase
- Step 3: Coherent - implementation phase
- Step 4: Coherent - completion

Internal Consistency: Notes port conversion to integer, consistent approach

model5_completeness

Problem Coverage:

- Flask app creation: Yes
- SMTP configuration: Yes
- Template folder: Yes
- Mail initialization: Yes
- Test route: Yes
- Return app: Yes

Factual Correctness: Correctly identifies need for port integer conversion

Constraint and edge case handling:

- Port type conversion
 - Placeholder recipient
 - Basic implementation
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Model 6 Evaluation

model6_efficiency

Total steps: 6 (plus complete code implementation)

- Step 1: Understanding requirements - **Essential**
- Step 2: Create Flask app - **Essential**
- Step 3: Configure SMTP - **Essential**
- Step 4: Initialize Mail - **Essential**
- Step 5: Create test route - **Essential**
- Step 6: Return app - **Essential**

model6_logic

Step Coherence:

- Step 1: Starting point
- Step 2: Coherent - app with template folder
- Step 3: Coherent - email configuration
- Step 4: Coherent - Mail object creation
- Step 5: Coherent - test functionality
- Step 6: Coherent - return requirement

Internal Consistency: No contradictions, includes working code

model6_completeness

Problem Coverage:

- Flask app creation: Yes
- SMTP configuration: Yes
- Template folder: Yes
- Mail initialization: Yes
- Test route: Yes (with error handling)
- Return app: Yes
- Complete working code: Yes

Factual Correctness: All correct, includes both text and HTML email body

Constraint and edge case handling:

- Try-except error handling
- Both text and HTML email bodies
- TLS enabled by default
- Comprehensive implementation

Summary

- **Most Complete:** Model 6 (includes full working code with error handling)
- **Most Efficient:** Model 5 (only 4 concise steps)
- **Best Logic:** Models 3 and 4 (smart port-based security selection)
- **Most Practical:** Model 6 (ready-to-use implementation)